

Data Center Performance Project

Simulation and Operational Analysis of a Technological Plant

PostgreSQL	Python · Pandas	Power BI	SQLAlchemy
------------	-----------------	----------	------------

Objective

The project simulates an operational monitoring system for an industrial data center, covering the full data pipeline lifecycle: from relational database modeling to operational KPI visualization. The data is built using real-world logic (seasonality, hourly workloads, incident distribution)

Pipeline

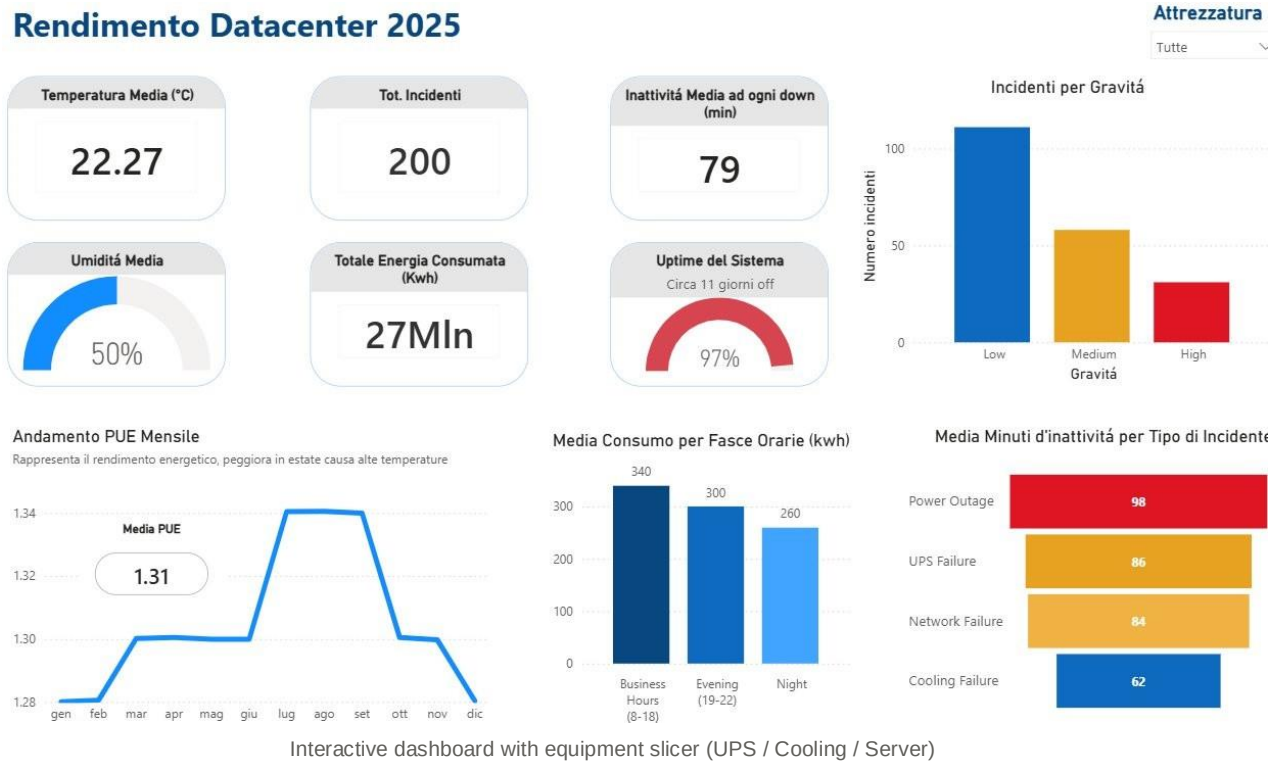
01 Database	02 Python	03 Power BI
Star schema in PostgreSQL. 5 tables: 3 dimensional (equipment, location, date) and 2 fact tables (metrics, incidents).	Synthetic data generation with NumPy and Pandas. DB loading via SQLAlchemy. EDA with Matplotlib and Seaborn.	Interactive dashboard with equipment slicer. Operational KPIs, seasonal PUE analysis, incidents by severity and type.

Database Schema

Table	Type	Key Columns
dim_equipment	Dimensional	equipment_id, equipment_name, equipment_type, manufacturer
dim_location	Dimensional	location_id, room_name, floor
dim_date	Dimensional	date_id, full_date, day, month, quarter, year
fact_metrics	Fact	timestamp, equipment_id, location_id, temperature, humidity, power_kw, pue, ups_efficiency
fact_incidents	Fact	timestamp, equipment_id, incident_type, severity, downtime_minutes

Dashboard Power BI

Rendimento Datacenter 2025



Key Insight

Seasonal PUE	The annual average PUE is 1.31. During summer months (Jul–Sep) it rises to 1.34 due to higher ambient temperatures, with a direct impact on cooling system efficiency.
Consumption by time slot	Average consumption during business hours (8–18) is 340 kWh, 31% higher than at night (260 kWh), consistent with peak server utilization during business activity.
Critical Incidents	Power Outage and UPS Failure generate the longest downtimes (98 and 86 min on average). These are priority areas for preventive maintenance, along with Cooling Failure (35% of total events)
System Uptime	The system records 97% uptime annually — approximately 11 days of total inactivity — a figure that should be reduced. N.B. The incident count was generated entirely at random and has no correlation with temperature or other metrics.

[Link file python](#)

[Link file sql](#)

[Link dashboard PowerBi](#)